NANE NARIMANYAN

MARKETING ANALYTICS

HOMEWORK 1

CITIZENS OF HUMANITY × PILI ECO-INDIGO

1. Selected innovation

I chose Citizens of Humanity × Pili Eco-Indigo from TIME's Best Inventions of 2025, in the Fashion category. Eco-Indigo is a bio-based dye developed as an alternative to conventional petroleum-derived indigo. It can be used to dye denim while reducing dependence on fossil-based raw materials. The first commercial denim products were introduced in 2025 [1–3].

2. Similar innovation from the past

For the historical comparison, I chose sustainable and organic cotton production. Both involve replacing conventional inputs or practices in the textile industry with alternatives intended to lower environmental impact. Their adoption depends on whether producers can make the switch, the cost, and the interest of brands and customers. There is also an important difference: sustainable cotton concerns farming and sourcing, while Eco-Indigo concerns dye production. I use cotton as a comparison for industry adoption, not as an identical technology.

3. Historical data

I used the OECD-FAO Agricultural Outlook 2022–2031, Cotton chapter, Table 10.1 [4]. It reports the share of global cotton production classified as sustainable or organic from 2010 to 2018. These values measure a share of production, not the number of companies adopting a product. I used the published yearly percentages for the model.

Year	Observed share (%)	Bass fitted share (%)
2010	1	1.06
2011	2	2.49
2012	5	4.35
2013	6	6.72
2014	9	9.61
2015	15	13.00
2016	16	16.76
2017	20	20.70
2018	25	24.58

4. Estimating the Bass Model

The Bass Model describes adoption through innovation (p), which represents independent decisions to adopt, and imitation (q), which represents adoption influenced by earlier adopters. The cumulative fraction

of potential adopters is F(t) = [1 − exp(−(p+q)t)] / [1 + (q/p)exp(−(p+q)t)]. I fitted this curve to the historical cotton-production shares using constrained nonlinear least squares, treating 2010 as t = 1. This starting point is an assumption, since sustainable cotton was already in use before 2010. The resulting estimates are p = 0.02158, q = 0.33026, and M = 0.42215 (42.22% of world cotton production). The in-sample root mean squared error is about 0.87 percentage points. Since q is larger than p, the fitted curve is more strongly influenced by the imitation term. The 42.22% is a production share for the historical comparison, not the number of Eco-Indigo customers.

5. Predicting Eco-Indigo adoption

I used the historical estimates of p and q to create a possible adoption path for Eco-Indigo beginning in 2025. This assumes that Eco-Indigo spreads at a similar pace to the historical comparison. I did not transfer the cotton market potential because the two measures are different. Instead, I defined an Eco-Indigo adopter as a textile organization or brand that begins using the dye. The projected curve increases slowly at first, grows more quickly as adoption spreads, and eventually levels off.

6. Geographical scope

I chose a global scope because Eco-Indigo is part of an international denim supply chain, involving partners in more than one country [2]. The historical comparison also uses worldwide cotton-production data [4]. A global approach is therefore more consistent with the data and the product’s potential market than choosing a single country.

7. Number of adopters and assumptions

Since I could not find a reliable public count of all organizations that might adopt Eco-Indigo, I used a Fermi estimate for the market size: 20 possible denim-producing markets × 50 potential organizations in each = 1,000 organizations. These two figures are assumptions made for the scenario, not verified market statistics. The forecast uses A(t) = 1,000 × F(t) for cumulative adopters, and annual new adopters are calculated as the difference between consecutive cumulative totals. For example, the model gives about 308 cumulative adopters in 2030. The largest annual increase is approximately 93 organizations in 2032.

Year	New adopters	Total adopters	Adoption (%)
2025	25	25	2.52
2026	34	59	5.89
2027	44	103	10.3
2028	56	159	15.91
2029	69	228	22.77
2030	80	308	30.8
2031	89	397	39.7
2032	93	490	49.03
2033	92	582	58.22
2034	85	667	66.74
2035	75	742	74.22

Year	New adopters	Total adopters	Adoption (%)
2036	62	805	80.46
2037	50	855	85.47
2038	39	893	89.35
2039	29	923	92.28
2040	22	944	94.45

8. Sensitivity and limitations

I also considered markets of 500 and 2,000 organizations. With the same p and q, the projected number of adopters in 2040 would be about 472 and 1,889, respectively, compared with 944 in the base scenario of 1,000. The main limitation is that historical cotton production shares are not the same as counts of companies adopting a new dye. The historical period is short, the starting year is assumed, and the Eco-Indigo market size is hypothetical. These results should therefore be read as possible scenarios, not confirmed sales or actual customer counts.

Conclusion

The Bass Model provides a way to explore how Eco-Indigo might spread through the denim industry. Based on the historical textile-sector comparison, the model suggests that adoption could accelerate during the early 2030s and then slow as the potential market fills. The forecast is useful for illustrating the diffusion process, but a stronger prediction would require actual data on Eco-Indigo customers and production capacity.

References

[1] TIME (2025). Citizens of Humanity and Pili Eco-Indigo. https://time.com/collections/best-inventions-2025/7318322/citizens-of-humanity-and-pili-eco-indigo/?filters=Fashion

[2] Pili (2025). Pili Partners with Citizens of Humanity and Orta. https://www.pili.bio/en/pili-partners-with-citizens-of-humanity-and-orta-to-revolutionize-the-denim-industry-with-biobased-indigo/

[3] Citizens of Humanity. ECO-INDIGO. https://citizensofhumanity.com/pages/eco-indigo

[4] OECD-FAO (2022). Agricultural Outlook 2022–2031, Cotton, Table 10.1. https://www.oecd.org/en/publications/oecd-fao-agricultural-outlook-2022-2031_f1b0b29c-en/full-report/component-13.html

[5] Hovhannisyan, K. DS 223 Marketing Analytics: Bass Model for Innovation Diffusion. Course lecture.